

Lab 3: Architecture Justification

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Architecture selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System. Presentation contains User Interface; application/business contains Order Manager; infrastructure/data contains Payment Service, Receipt Printer, and Menu Repository. These are logical layers in one kiosk application.

Reason 1: Separation of concerns

The touchscreen offers Espresso, Americano, and Latte in Small or Large sizes. Keeping display logic separate from order validation and pricing allows screen changes without changing the checkout rules. UI requires Order Interface instead of accessing storage or drivers directly.

Reason 2: Replaceable adapters

Payment Service encapsulates credit-card authorization, Receipt Printer encapsulates the printer driver, and Menu Repository encapsulates menu/price storage. Their explicit contracts let a device or storage adapter change while Order Manager keeps its workflow.

Security advantage

A card-terminal SDK captures and authorizes card data. Payment Interface returns the outcome and transaction reference to Order Manager; raw card data is excluded from storage and receipts. This limits exposure, although logical layering alone does not provide process isolation or establish payment compliance.

Performance benefit

Local API calls and cached menu prices avoid remote round trips during selection and total calculation. Payment authorization may still need a network call. The design aims to keep the touch interface responsive; measured latency depends on implementation and testing.

Interfaces and interaction

The diagram shows five components and four ball/socket assemblies: Order, Payment, Receipt, and Menu. Order Manager validates the selection, reads pricing, requests payment, and prints only after approval. A print retry must not charge the customer again.